



Overview of Rocket Design and Structure

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1 Introduction

In this article, a brief overview of the Rocket design will be given. The dimensions and materials used for each component will be discussed. The rocket has a total height of 2.2 meters and weight of around 30 kgs. It is expected to reach an apogee of around 1 km.

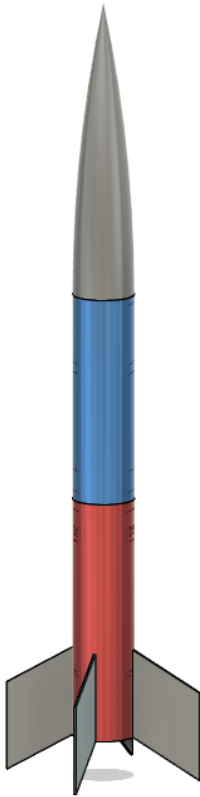


Figure 1: The Entire Rocket Assembly

2 Nosecone

The nosecone shape is a Tangent Ogive. The base diameter is 6 inches and its height is 30 inches. Its thickness is 10 mm. This particular shape has been chosen as it is very common in amateur rocketry and the drag provided by it is quite small. It has been decided to manufacture it by 3D-printing it in parts and attaching them by means of joints or by constructing a framework. The hollow space inside the nosecone will be used to store the parachute. It will be friction-fitted and will come off during parachute deployment. The deployment is expected to occur by means of a small explosion.



Figure 2: Nosecone

3 Avionics Bay

The Avionic bay shall contain two Raspberry Pi micorprocessors, an accelerometer, a gyroscope, GPS, barometer and chamber pressure sensor along with a primary battery and backup battery. They will be housed in an aluminium structure consisting of two plates and four rods. The structure will be fitted using screws inside a tube made of PVC pipe of diameter 6 inches and around 21 inches. The lowermost aluminium plate will be joined to a connector tube which joins the avionics bay to the forward closure of motor tube.



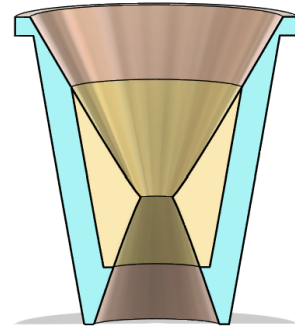
Figure 3: Avionics Bay

4 Motor

The motor consists of the two fuel grains (composed of KNSB), the nozzle and the motor tube. This assembly is joined to the rest of the rocket through the forward closure mentioned above. The nozzle diameter is 6 inches and height is 6.5 inches. The convergence angle is 30 degrees and divergence angle is 18 degrees. The nozzle is made of graphite and insulated by a layer of phenolic resin. The nozzle carrier is composed of aluminium. The motor tube of height 24 inches and diameter 6 inches is also made of aluminium. Between the grains and motor tube, there is a cardboard tube for insulation and separation and it will be consumed during combustion.



(a) Internal Structure



(b) Cross Section

Figure 4: Motor

5 Fins

The fins are important for the stability of the rocket and helping it maintain a straight trajectory. They are to be 3D-printed out of PLA and have the geometry of a parallelogram with base and height of 9 inches and base angle of 60 degrees. The dimensions are not final and subject to change.

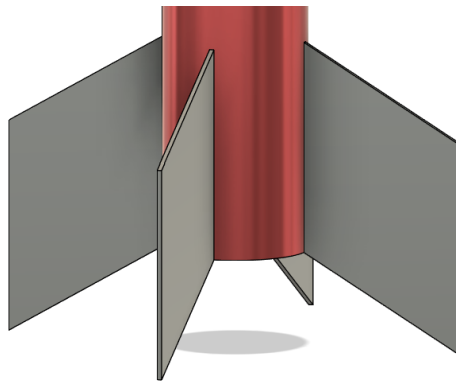


Figure 5: Fins

6 Conclusion

A proposal has been made to manufacture the rocket in the manner discussed and lay the foundations of IIT Guwahati's rocketry team. With proper funding and guidance, we are determined to create a culture of competitive rocketry in IIT Guwahati which has been prevalent in many other IITs.